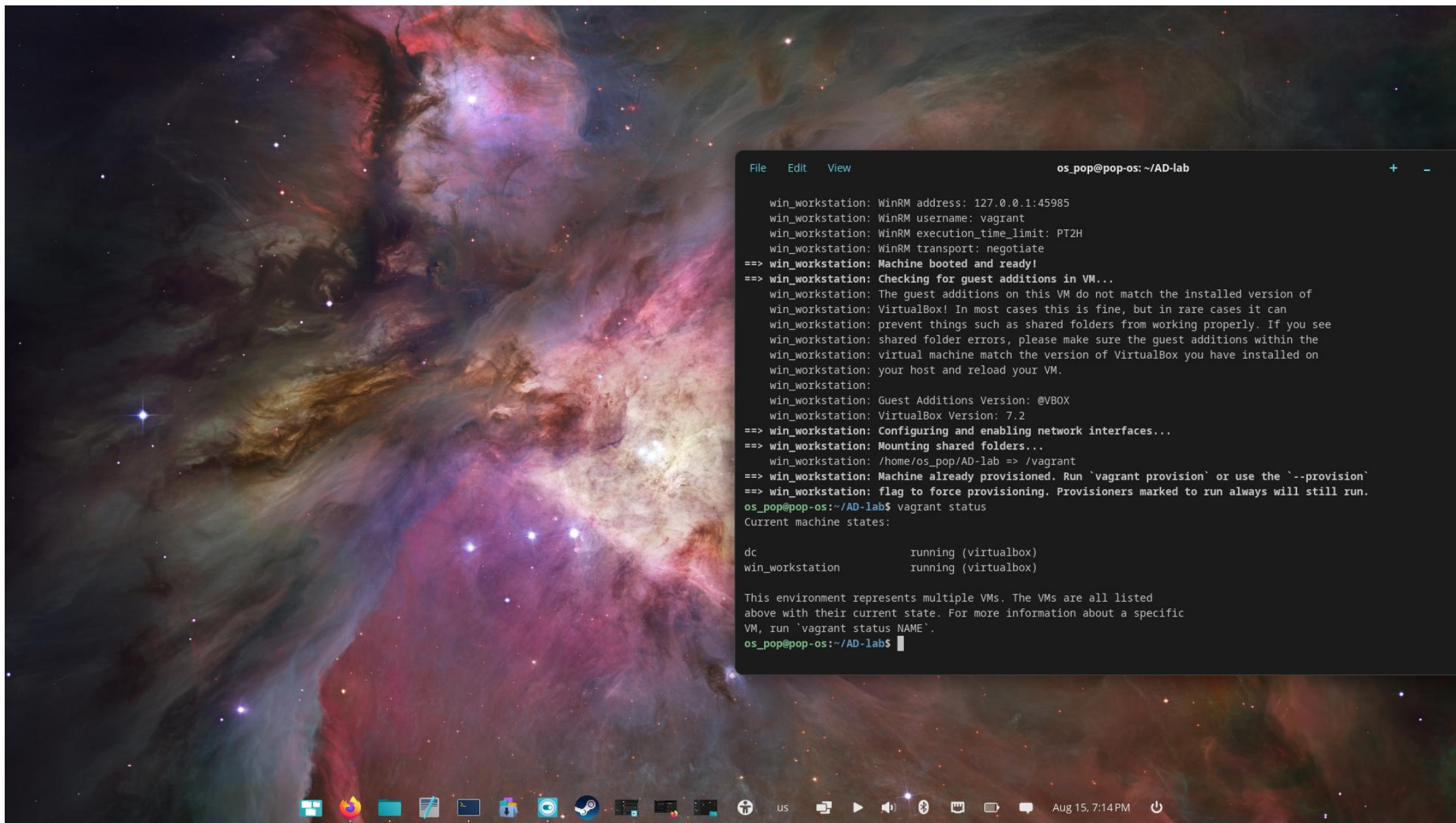




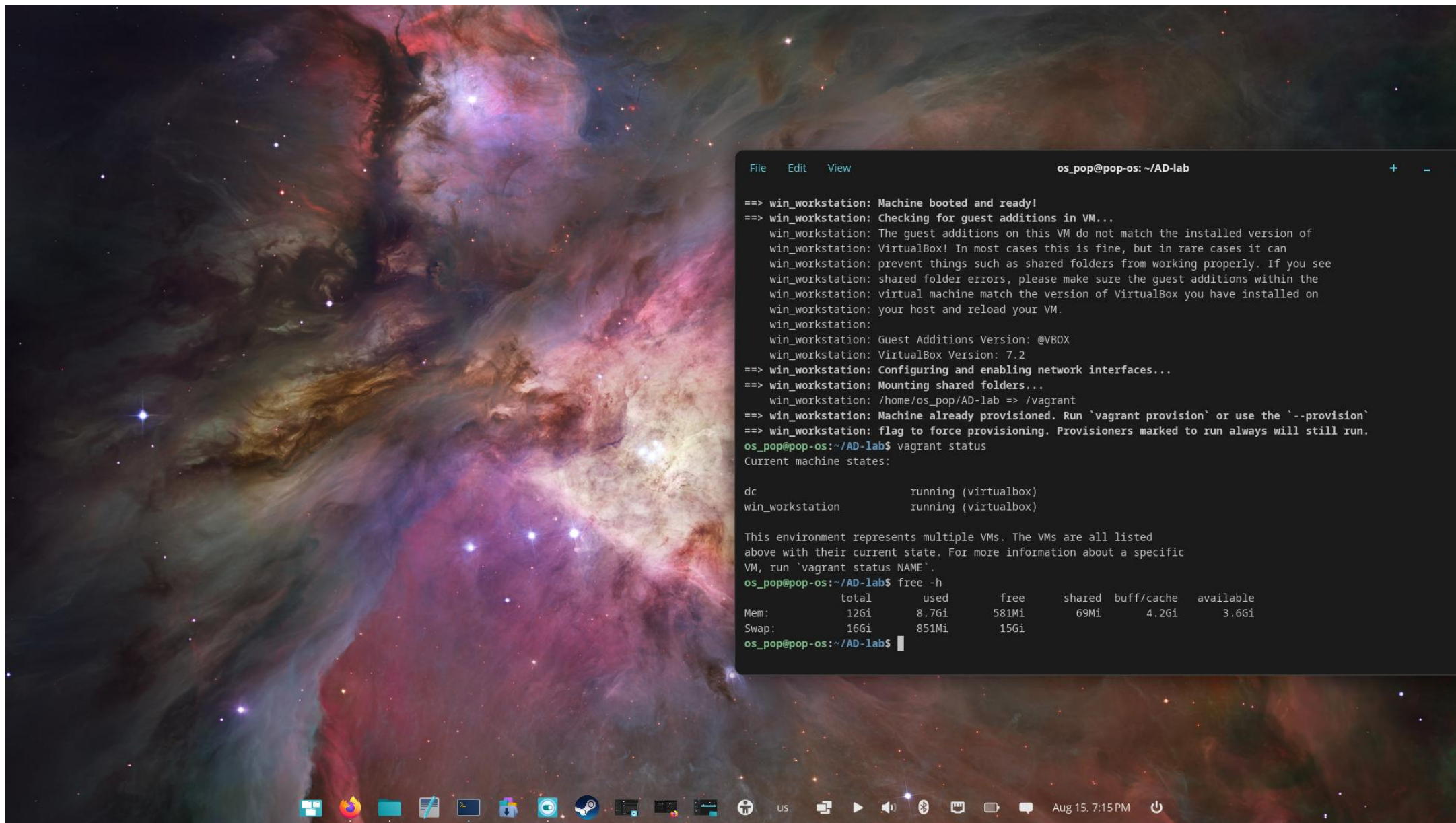
File Edit View

os_pop@pop-os: ~/AD-lab

```
win_workstation: WinRM address: 127.0.0.1:45985
win_workstation: WinRM username: vagrant
win_workstation: WinRM execution_time_limit: PT2H
win_workstation: WinRM transport: negotiate
==> win_workstation: Machine booted and ready!
==> win_workstation: Checking for guest additions in VM...
win_workstation: The guest additions on this VM do not match the installed version of
win_workstation: VirtualBox! In most cases this is fine, but in rare cases it can
win_workstation: prevent things such as shared folders from working properly. If you see
win_workstation: shared folder errors, please make sure the guest additions within the
win_workstation: virtual machine match the version of VirtualBox you have installed on
win_workstation: your host and reload your VM.
win_workstation: Guest Additions Version: @VBOX
win_workstation: VirtualBox Version: 7.2
==> win_workstation: Configuring and enabling network interfaces...
==> win_workstation: Mounting shared folders...
win_workstation: /home/os_pop/AD-lab => /vagrant
==> win_workstation: Machine already provisioned. Run `vagrant provision` or use the `--provision`
==> win_workstation: flag to force provisioning. Provisioners marked to run always will still run.
os_pop@pop-os:~/AD-lab$ vagrant status
Current machine states:

dc                           running (virtualbox)
win_workstation              running (virtualbox)

This environment represents multiple VMs. The VMs are all listed
above with their current state. For more information about a specific
VM, run `vagrant status NAME`.
os_pop@pop-os:~/AD-lab$
```



File Edit View

os_pop@pop-os: ~/AD-lab

```
==> win_workstation: Machine booted and ready!
==> win_workstation: Checking for guest additions in VM...
win_workstation: The guest additions on this VM do not match the installed version of
win_workstation: VirtualBox! In most cases this is fine, but in rare cases it can
win_workstation: prevent things such as shared folders from working properly. If you see
win_workstation: shared folder errors, please make sure the guest additions within the
win_workstation: virtual machine match the version of VirtualBox you have installed on
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win_workstation:
win_workstation: Guest Additions Version: @VBOX
win_workstation: VirtualBox Version: 7.2
==> win_workstation: Configuring and enabling network interfaces...
==> win_workstation: Mounting shared folders...
win_workstation: /home/os_pop/AD-lab => /vagrant
==> win_workstation: Machine already provisioned. Run `vagrant provision` or use the `--provision`
==> win_workstation: flag to force provisioning. Provisioners marked to run always will still run.
os_pop@pop-os:~/AD-lab$ vagrant status
Current machine states:
```

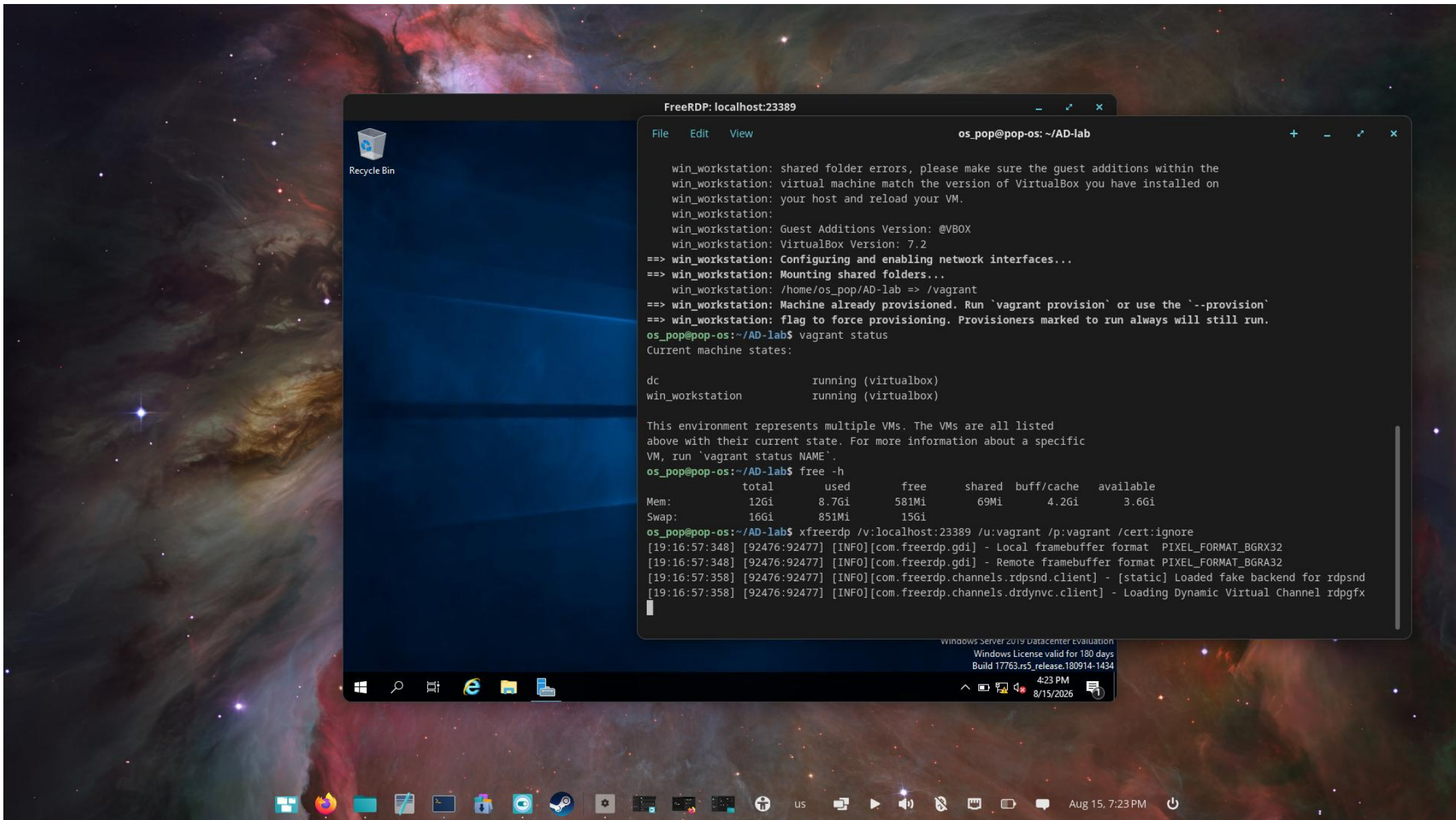
```
dc                running (virtualbox)
win_workstation   running (virtualbox)
```

This environment represents multiple VMs. The VMs are all listed above with their current state. For more information about a specific VM, run `vagrant status NAME`.

```
os_pop@pop-os:~/AD-lab$ free -h
```

	total	used	free	shared	buff/cache	available
Mem:	12Gi	8.7Gi	581Mi	69Mi	4.2Gi	3.6Gi
Swap:	16Gi	851Mi	15Gi			

```
os_pop@pop-os:~/AD-lab$
```



FreeRDP: localhost:23389

File Edit View

os_pop@pop-os: ~/AD-lab

```
win_workstation: shared folder errors, please make sure the guest additions within the
win_workstation: virtual machine match the version of VirtualBox you have installed on
win_workstation: your host and reload your VM.
win_workstation:
win_workstation: Guest Additions Version: @VBOX
win_workstation: VirtualBox Version: 7.2
==> win_workstation: Configuring and enabling network interfaces...
==> win_workstation: Mounting shared folders...
win_workstation: /home/os_pop/AD-lab => /vagrant
==> win_workstation: Machine already provisioned. Run `vagrant provision` or use the `--provision`
==> win_workstation: flag to force provisioning. Provisioners marked to run always will still run.
os_pop@pop-os:~/AD-lab$ vagrant status
Current machine states:
```

```
dc                running (virtualbox)
win_workstation   running (virtualbox)
```

This environment represents multiple VMs. The VMs are all listed
above with their current state. For more information about a specific
VM, run `vagrant status NAME`.

```
os_pop@pop-os:~/AD-lab$ free -h
```

	total	used	free	shared	buff/cache	available
Mem:	12Gi	8.7Gi	581Mi	69Mi	4.2Gi	3.6Gi
Swap:	16Gi	851Mi	15Gi			

```
os_pop@pop-os:~/AD-lab$ xfreerdp /v:localhost:23389 /u:vagrant /p:vagrant /cert:ignore
```

```
[19:16:57:348] [92476:92477] [INFO][com.freerdp.gdi] - Local framebuffer format PIXEL_FORMAT_BGRX32
```

```
[19:16:57:348] [92476:92477] [INFO][com.freerdp.gdi] - Remote framebuffer format PIXEL_FORMAT_BGRA32
```

```
[19:16:57:358] [92476:92477] [INFO][com.freerdp.channels.rdpnd.client] - [static] Loaded fake backend for rdpnd
```

```
[19:16:57:358] [92476:92477] [INFO][com.freerdp.channels.drdynvc.client] - Loading Dynamic Virtual Channel rdpgfx
```

Windows Server 2019 Datacenter Evaluation

Windows License valid for 180 days

Build 17763.rs5_release.180914-1434

4:23 PM

8/15/2026

Aug 15, 7:23 PM

FreeRDP: localhost:23389

Server Manager

Server Manager ▸ Local Server

Dashboard
Local Server
All Servers
AD DS
DNS
File and Storage Services ▸

PROPERTIES
For domain-controller

Computer name	domain-controller	Last installed updates	Today 4:14 PM
Domain	cyberloop.local	Windows Update	Install updates
		Last checked for updates	Today 4:14 PM
Windows Defender Firewall	Domain: On	Windows Defender Antivirus	Real-time protection
Remote management	Enabled	Feedback & Diagnostics	Settings
Remote Desktop	Enabled	IE Enhanced Security Configuration	Off
NIC Teaming	Disabled	Time zone	(UTC-08:00) Pacific Time (US & Canada)
Ethernet	IPv4 address assigned by DHCP, IPv6 enabled	Product ID	00431-1000-8000-0000-0000-0000-0000-0000
Ethernet 2	192.168.56.10, IPv6 enabled		
Operating system version	Microsoft Windows Server 2019 Datacenter Evaluation	Processors	AMD Ryzen 7 5800X
Hardware information	innotek GmbH VirtualBox	Installed memory (RAM)	1.5 GB
		Total disk space	500 GB

TASKS

EVENTS
All events | 49 total

Filter

Server Name	ID	Severity	Source	Log	Date and Time
DOMAIN-CONTROLL	6008	Error	EventLog	System	8/15/2026 4:14:07 PM
DOMAIN-CONTROLL	34	Warning	Disk	System	8/15/2026 4:14:05 PM
DOMAIN-CONTROLL	34	Warning	Disk	System	8/15/2026 4:14:05 PM

TASKS

4:21 PM
8/15/2026

```
os: ~/AD-lab

the guest additions within the
VirtualBox you have installed on

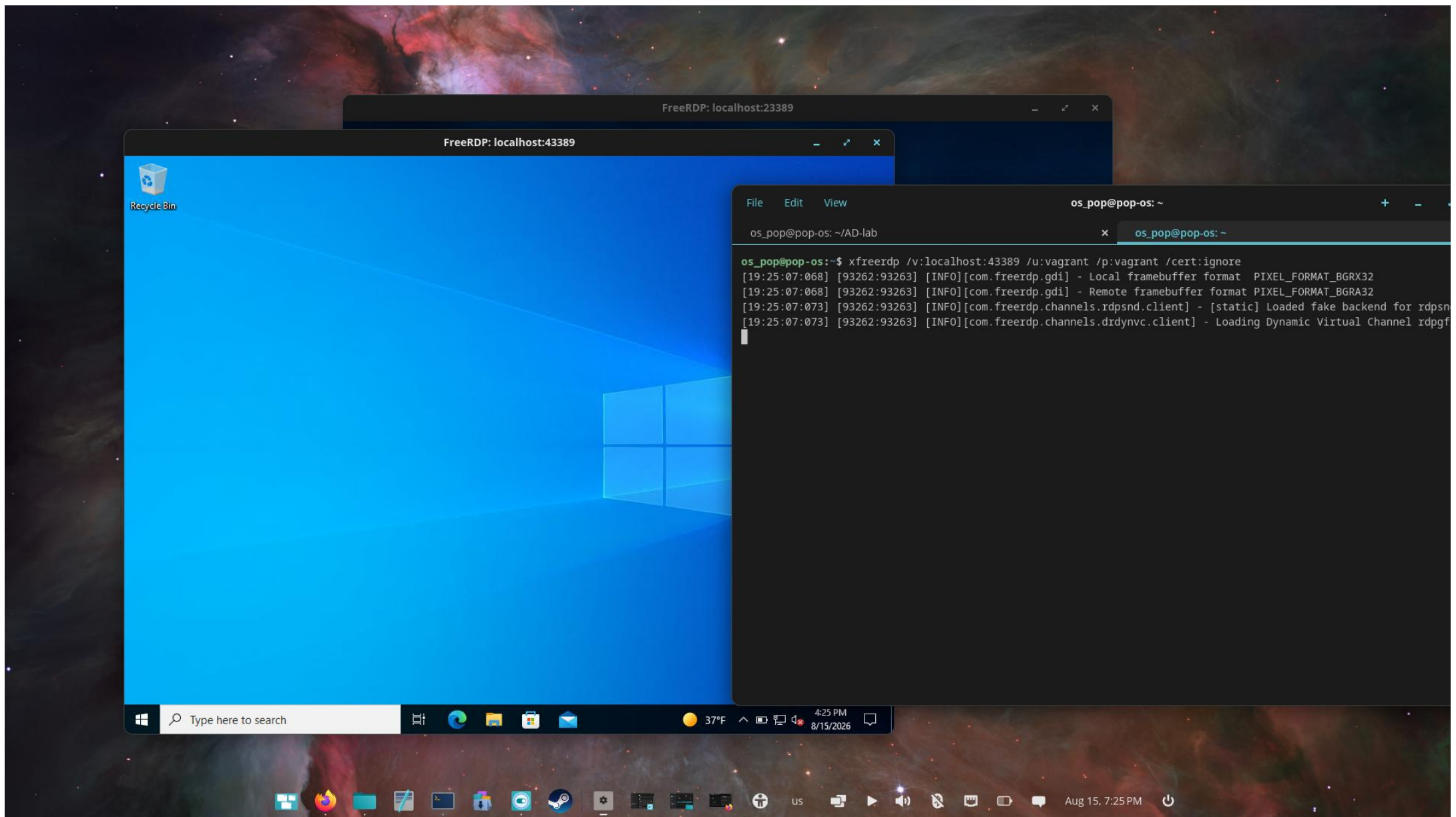
Faces...

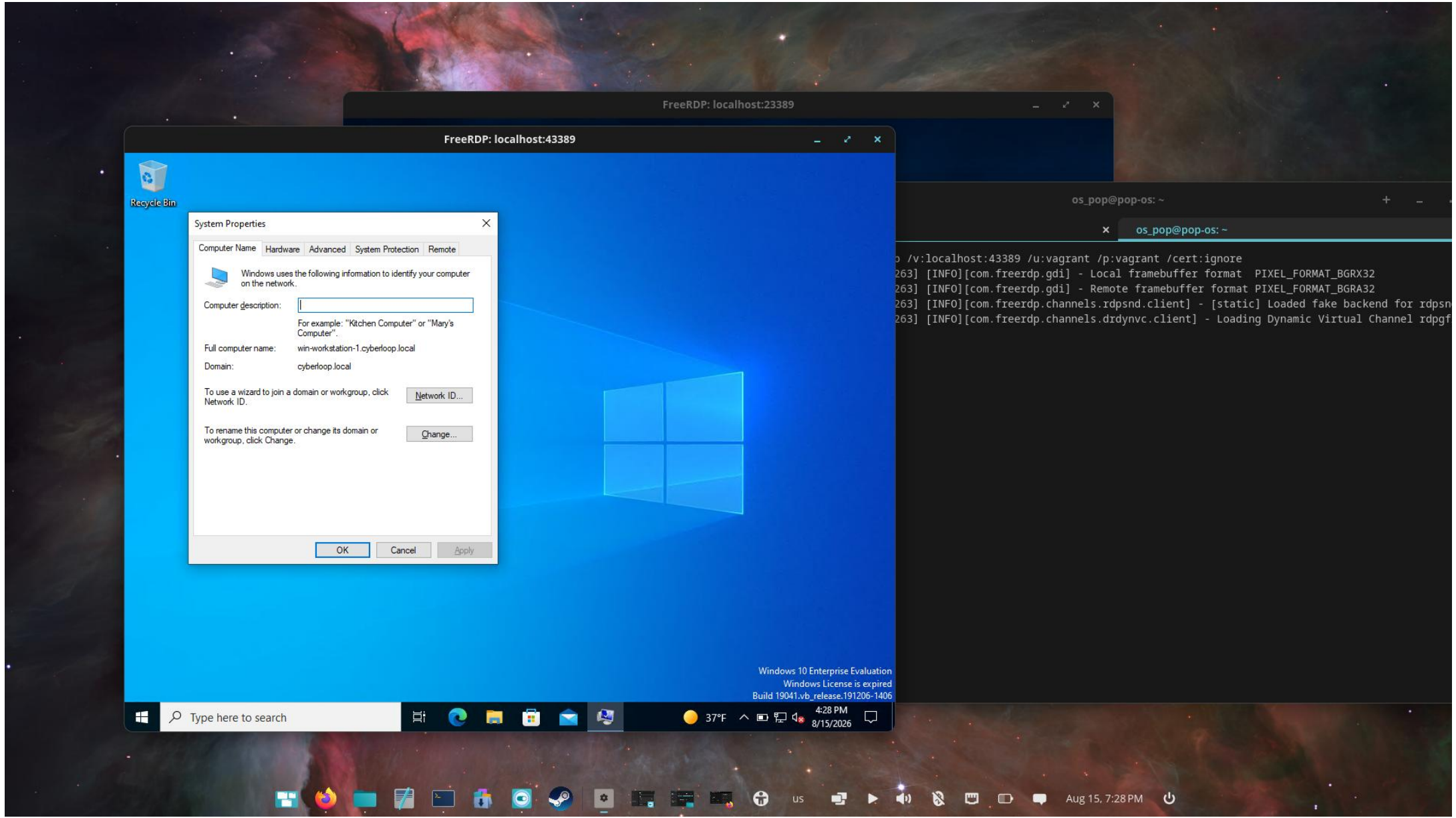
ant provision' or use the '--provision'
ers marked to run always will still run.

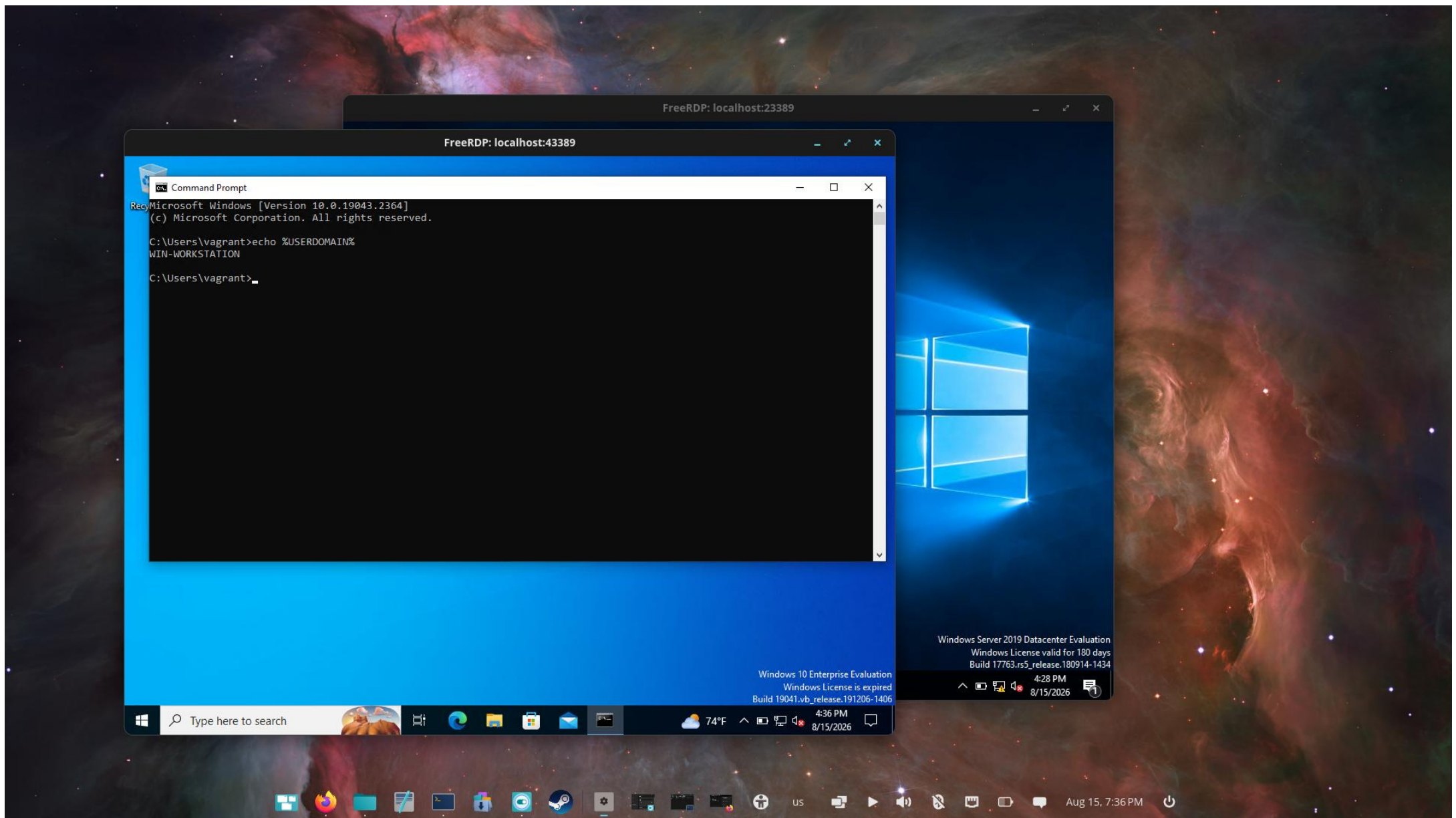
isted
a specific

uff/cache available
4.2Gi 3.6Gi

ant /p:vagrant /cert:ignore
l framebuffer format PIXEL_FORMAT_BGRX32
te framebuffer format PIXEL_FORMAT_BGRA32
psnd.client] - [static] Loaded fake backend for rdpsnd
dynvc.client] - Loading Dynamic Virtual Channel rdpgfx
```







Command Prompt

```
Microsoft Windows [Version 10.0.17134.1]
(c) Microsoft Corporation. All rights reserved.

C:\Users\vagrant>echo %USERDOMAIN%
WIN-WORKSTATION

C:\Users\vagrant>
```

FreeRDP: localhost:23389

Recycle Bin

Active Directory Users and Computers

File Action View Help

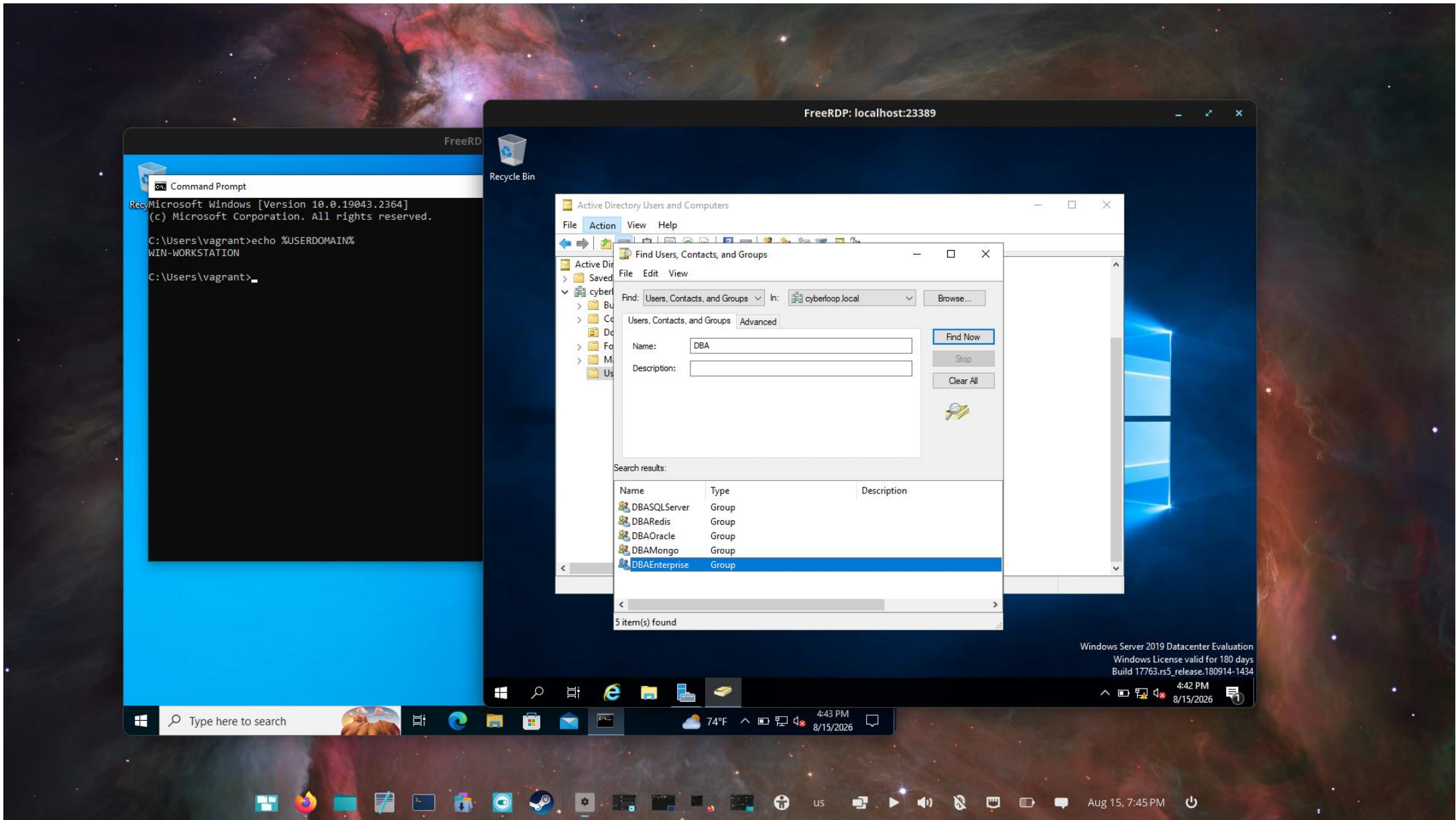
Active Directory Users and Computers

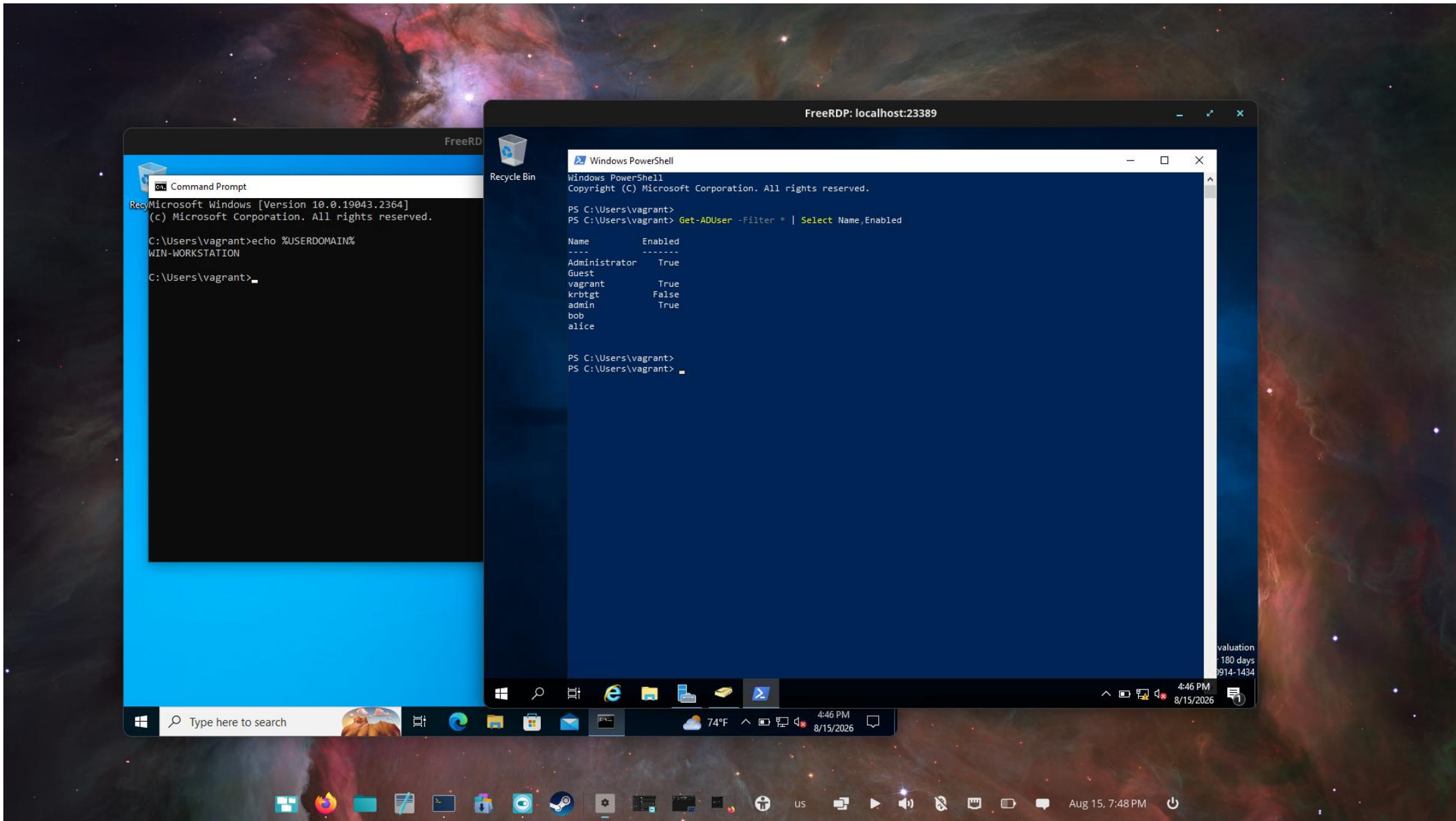
- Saved Queries
- cyberloop.local
 - Builtin
 - Computers
 - Domain Controllers
 - ForeignSecurityPrincipals
 - Managed Service Accounts
 - Users**

Name	Type	Description
admin	User	
Administrator	User	Built-in account for ad...
alice	User	
Allowed ROD...	Security Group...	Members in this group c...
bob	User	
Cert Publish...	Security Group...	Members of this group ...
Cloneable D...	Security Group...	Members of this group t...
Denied ROD...	Security Group...	Members in this group c...
DnsAdmins	Security Group...	DNS Administrators Gro...
DnsUpdateP...	Security Group...	DNS clients who are per...
Domain Ad...	Security Group...	Designated administrato...
Domain Co...	Security Group...	All workstations and ser...
Domain Con...	Security Group...	All domain controllers i...
Domain Gue...	Security Group...	All domain guests
Domain Users	Security Group...	All domain users
Enterprise A...	Security Group...	Designated administrato...
Enterprise K...	Security Group...	Members of this group ...
Enterprise R...	Security Group...	Members of this group ...
Group Polic...	Security Group...	Members in this group c...
Guest	User	Built-in account for gue...
Key Admins	Security Group...	Members of this group ...

Windows Server 2019 Datacenter Evaluation
Windows License valid for 180 days
Build 17763.rs5_release.180914-1434

4:31 PM
8/15/2026





FreeRD

Command Prompt

Microsoft Windows [Version 10.0.19043.2364]
(c) Microsoft Corporation. All rights reserved.

C:\Users\vagrant>echo %USERDOMAIN%
WIN-WORKSTATION

C:\Users\vagrant>

FreeRD: localhost:23389

Server Manager

Server Manager ▸ Local Server

Active Directory Users and Computers

File Action View Help

Local Server

Active Directory Users and Computers

Name	Type	Description
Builtin	builtinDomain	
Computers	Container	Default container for up...
Domain Con...	Organizational...	Default container for do...
ForeignSecu...	Container	Default container for sec...
IT Support	Organizational...	
Managed Se...	Container	Default container for ma...
Users	Container	Default container for up...
Contractors	Organizational...	
Hospital Staff	Organizational...	
allTeams	Security Group...	
DBAEnterprise	Security Group...	
DBAMongo	Security Group...	
DBAOracle	Security Group...	
DBARedis	Security Group...	
DBASQLServer	Security Group...	
JustATestDe...	Security Group...	
Keys	Unknown	

Server Name	ID	Severity	Source	Log	Date and Time
DOMAIN-CONTROLL	6008	Error	EventLog	System	8/15/2026 4:14:07 PM
DOMAIN-CONTROLL	34	Warning	Disk	System	8/15/2026 4:14:05 PM
DOMAIN-CONTROLL	34	Warning	Disk	System	8/15/2026 4:14:05 PM

5:01 PM
8/15/2026

FreeRDP: localhost:23389

Server Manager

Server Manager ▸ Local Server

Dashboard

Local Server

Group Policy Management

Forest: cyberloop.local

Domains

cyberloop.local

Default Domain

Contractors

Domain Control

Hospital Staff

IT Support

IT Support - Group Policy Objects

WMI Filters

Starter GPOs

Sites

Group Policy Modeling

Group Policy Results

PROPERTIES

For domain-controller

IT Support

Linked Group Policy Objects

This list does not include any GPOs linked to sites. For more details, see Help.

Precedence	GPO	Location	GPO Status	WMI Filter
1	IT Support - Passwo...	IT Support	Enabled	None
2	Default Domain Policy	cyberloop.local	Enabled	None

DOMAIN-CONTROLL 6008 Error EventLog System 8/15/2026 4:14:07 PM

DOMAIN-CONTROLL 34 Warning Disk System 8/15/2026 4:14:05 PM

DOMAIN-CONTROLL 34 Warning Disk System 8/15/2026 4:14:05 PM

5:11 PM 8/15/2026

FreeRD

Command Prompt

Microsoft Windows [Version 10.0.19043.2364]
(c) Microsoft Corporation. All rights reserved.

C:\Users\vagrant>echo %USERDOMAIN%
WIN-WORKSTATION

C:\Users\vagrant>

FreeRD: localhost:23389

Server Manager

Server Manager ▸ Local Server

Active Directory Users and Computers

File Action View Help

Local Server

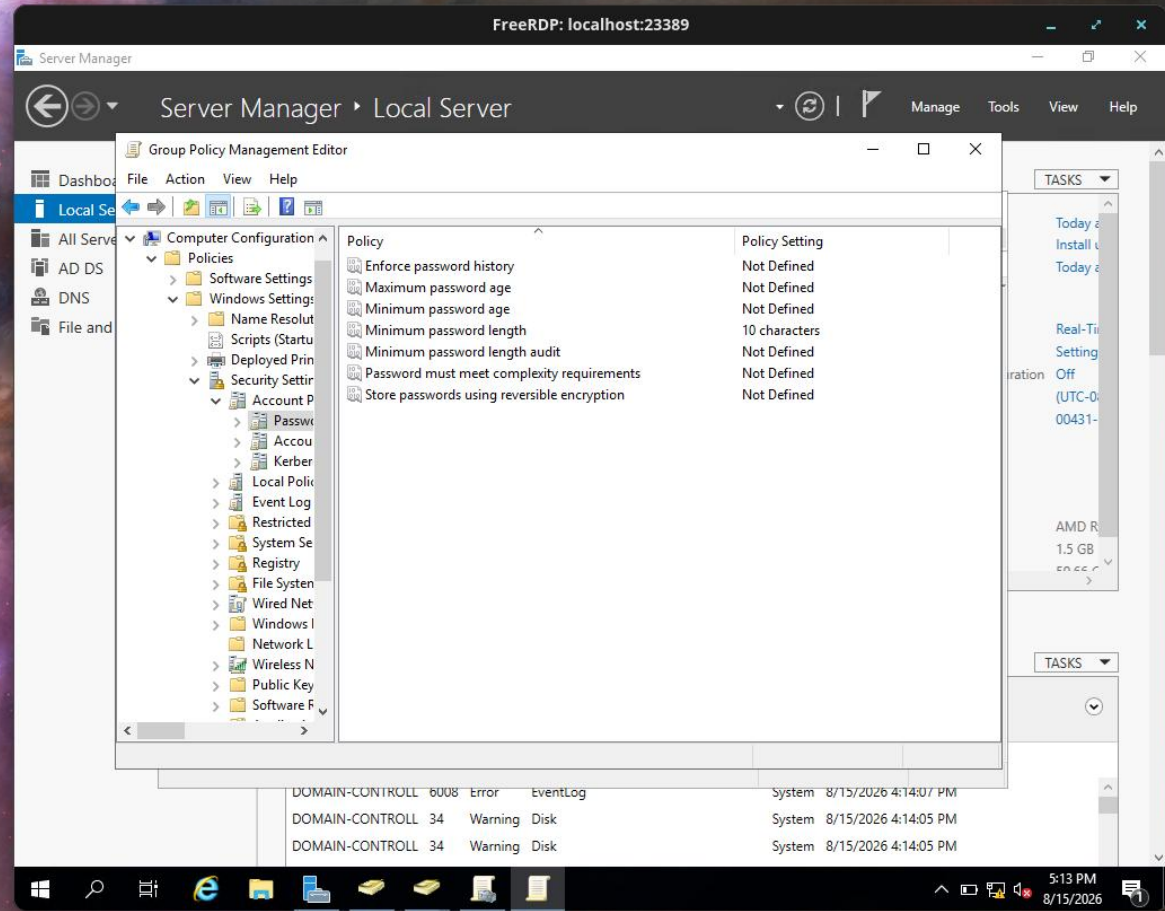
Active Directory Users and Computers

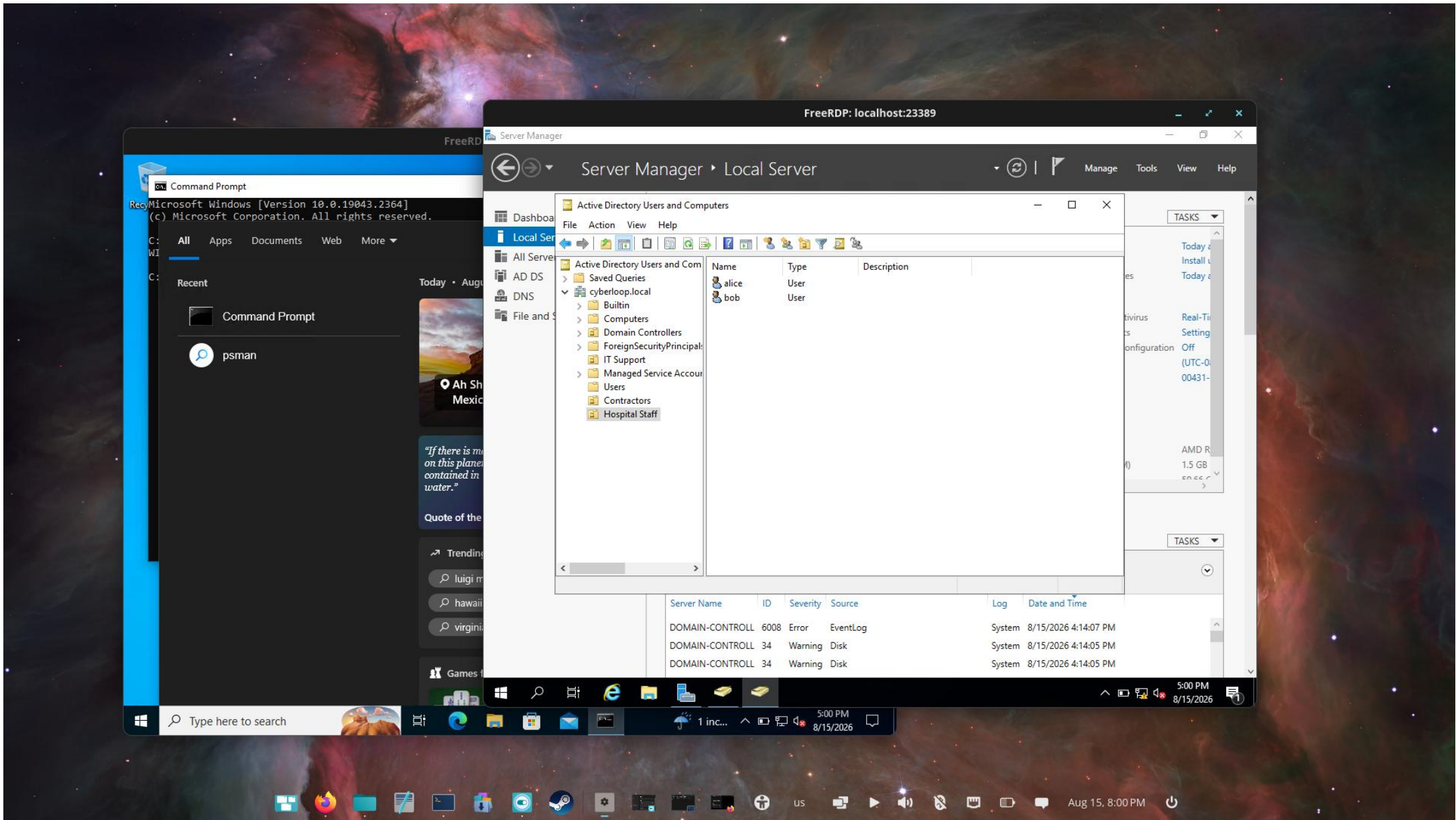
Name	Type	Description
IT Tech	User	

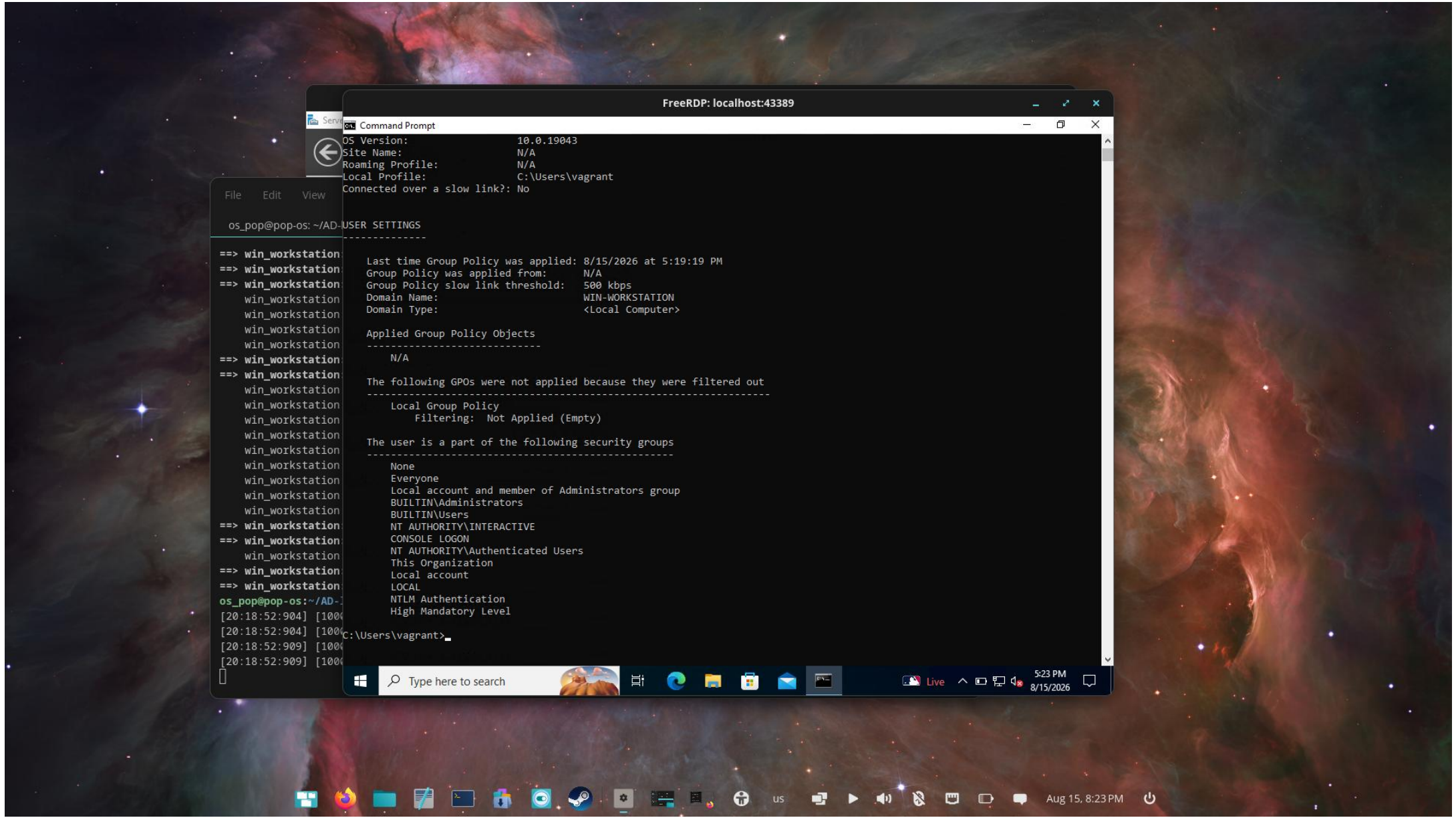
Server Name ID Severity Source Log Date and Time

DOMAIN-CONTROLL	6008	Error	EventLog	System	8/15/2026 4:14:07 PM
DOMAIN-CONTROLL	34	Warning	Disk	System	8/15/2026 4:14:05 PM
DOMAIN-CONTROLL	34	Warning	Disk	System	8/15/2026 4:14:05 PM

4:59 PM 8/15/2026







```
FreeRDP: localhost:43389
Command Prompt
OS Version: 10.0.19043
Site Name: N/A
Roaming Profile: N/A
Local Profile: C:\Users\vagrant
Connected over a slow link?: No

os_pop@pop-os: ~/AD-USER SETTINGS
=====
==> win_workstation
Last time Group Policy was applied: 8/15/2026 at 5:19:19 PM
==> win_workstation
Group Policy was applied from: N/A
==> win_workstation
Group Policy slow link threshold: 500 kbps
win_workstation
Domain Name: WIN-WORKSTATION
win_workstation
Domain Type: <Local Computer>
win_workstation
Applied Group Policy Objects
win_workstation
-----
N/A
==> win_workstation
The following GPOs were not applied because they were filtered out
win_workstation
-----
win_workstation
Local Group Policy
win_workstation
Filtering: Not Applied (Empty)
win_workstation
The user is a part of the following security groups
win_workstation
-----
win_workstation
None
win_workstation
Everyone
win_workstation
Local account and member of Administrators group
win_workstation
BUILTIN\Administrators
win_workstation
BUILTIN\Users
==> win_workstation
NT AUTHORITY\INTERACTIVE
==> win_workstation
CONSOLE LOGON
win_workstation
NT AUTHORITY\Authenticated Users
==> win_workstation
This Organization
win_workstation
Local account
==> win_workstation
LOCAL
os_pop@pop-os:~/AD-USER SETTINGS$
[20:18:52:904] [100%] C:\Users\vagrant>
```

FreeRD

Command Prompt

Microsoft Windows [Version 10.0.19043.2364]
(c) Microsoft Corporation. All rights reserved.

```
C:\Users\vagrant>echo %USERDOMAIN%  
WIN-WORKSTATION  
C:\Users\vagrant>
```

FreeRD: localhost:23389

Server Manager

Server Manager ▸ Local Server

Active Directory Users and Computers

File Action View Help

Local Server

Active Directory Users and Computers

Name	Type	Description
Contractor S...	User	

Server Name ID Severity Source Log Date and Time

DOMAIN-CONTROLL	6008	Error	EventLog	System	8/15/2026 4:14:07 PM
DOMAIN-CONTROLL	34	Warning	Disk	System	8/15/2026 4:14:05 PM
DOMAIN-CONTROLL	34	Warning	Disk	System	8/15/2026 4:14:05 PM

4:59 PM 8/15/2026

```
File Edit View
os_pop@pop-os: ~/AD-lab

==> win_workstation: Running 'pre-boot' VM c
==> win_workstation: Booting VM...
==> win_workstation: Waiting for machine to
win_workstation: WinRM address: 127.0.0.1
win_workstation: WinRM username: vagrant
win_workstation: WinRM execution_time_li
win_workstation: WinRM transport: negoti
==> win_workstation: Machine booted and read
==> win_workstation: Checking for guest addi
win_workstation: The guest additions on
win_workstation: VirtualBox! In most cas
win_workstation: prevent things such as
win_workstation: shared folder errors, p
win_workstation: virtual machine match t
win_workstation: your host and reload yo
win_workstation:
win_workstation: Guest Additions Version
win_workstation: VirtualBox Version: 7.2
==> win_workstation: Configuring and enablin
==> win_workstation: Mounting shared folders
win_workstation: /home/os_pop/AD-lab =>
==> win_workstation: Machine already provis
==> win_workstation: flag to force provision
os_pop@pop-os:~/AD-lab$ xfreerdp /v:localhos
[20:18:52:904] [100054:100055] [INFO][com.fr
[20:18:52:904] [100054:100055] [INFO][com.fr
[20:18:52:909] [100054:100055] [INFO][com.fr
[20:18:52:909] [100054:100055] [INFO][com.fr
```

```
Command Prompt
OS Version:
Site Name:
Roaming Profile:
Local Profile:
Connected over a s

USER SETTINGS
-----
Last time Group
Group Policy wa
Group Policy s
Domain Name:
Domain Type:

Applied Group
-----
N/A

The following
-----
Local Group
Filter:

The user is a
-----
None
Everyone
Local accou
BUILTIN\Adm
BUILTIN\Us
NT AUTHORI
CONSOLE LO
NT AUTHORI
This Organ
Local accou
LOCAL
NTLM Authent
High Mandatory Level

C:\Users\vagrant>
```

FreeRDP: localhost:23389

Server Manager

Group Policy Management

Group Policy Management Editor

File Action View Help

Computer Configuration

- Policies
- Software Settings
- Windows Settings
 - Name Resolut
 - Scripts (Startu
 - Deployed Prin
 - Security Setti
 - Account P
 - Passw
 - Accou
 - Kerber
 - Local Polic
 - Event Log
 - Restricted
 - System Se
 - Registry
 - File System
 - Wired Net
 - Windows I
 - Network L
 - Wireless N
 - Public Key
 - Software F

Policy

Policy Setting

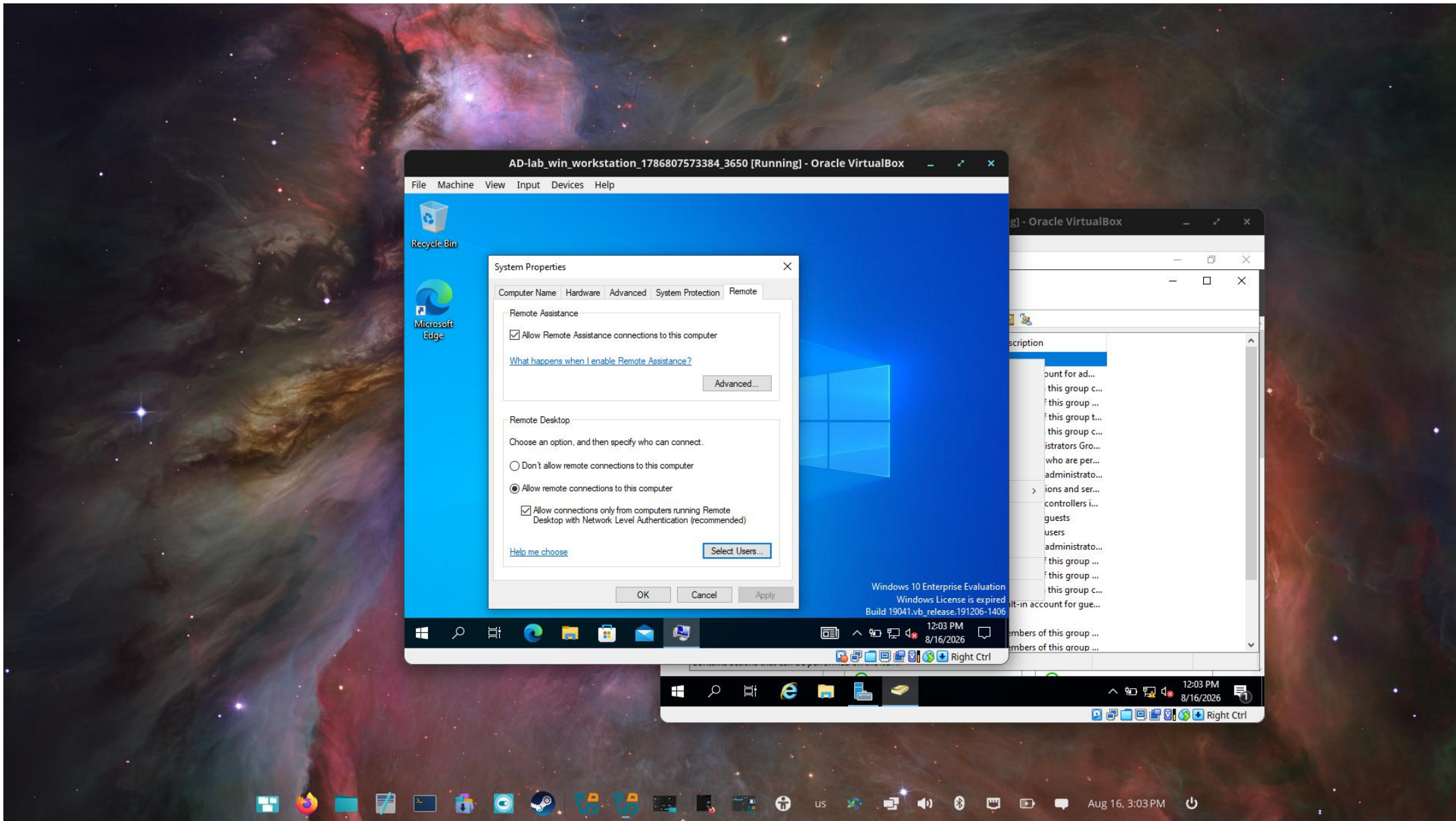
Account lockout duration	30 minutes
Account lockout threshold	3 invalid logon attempts
Reset account lockout counter after	30 minutes

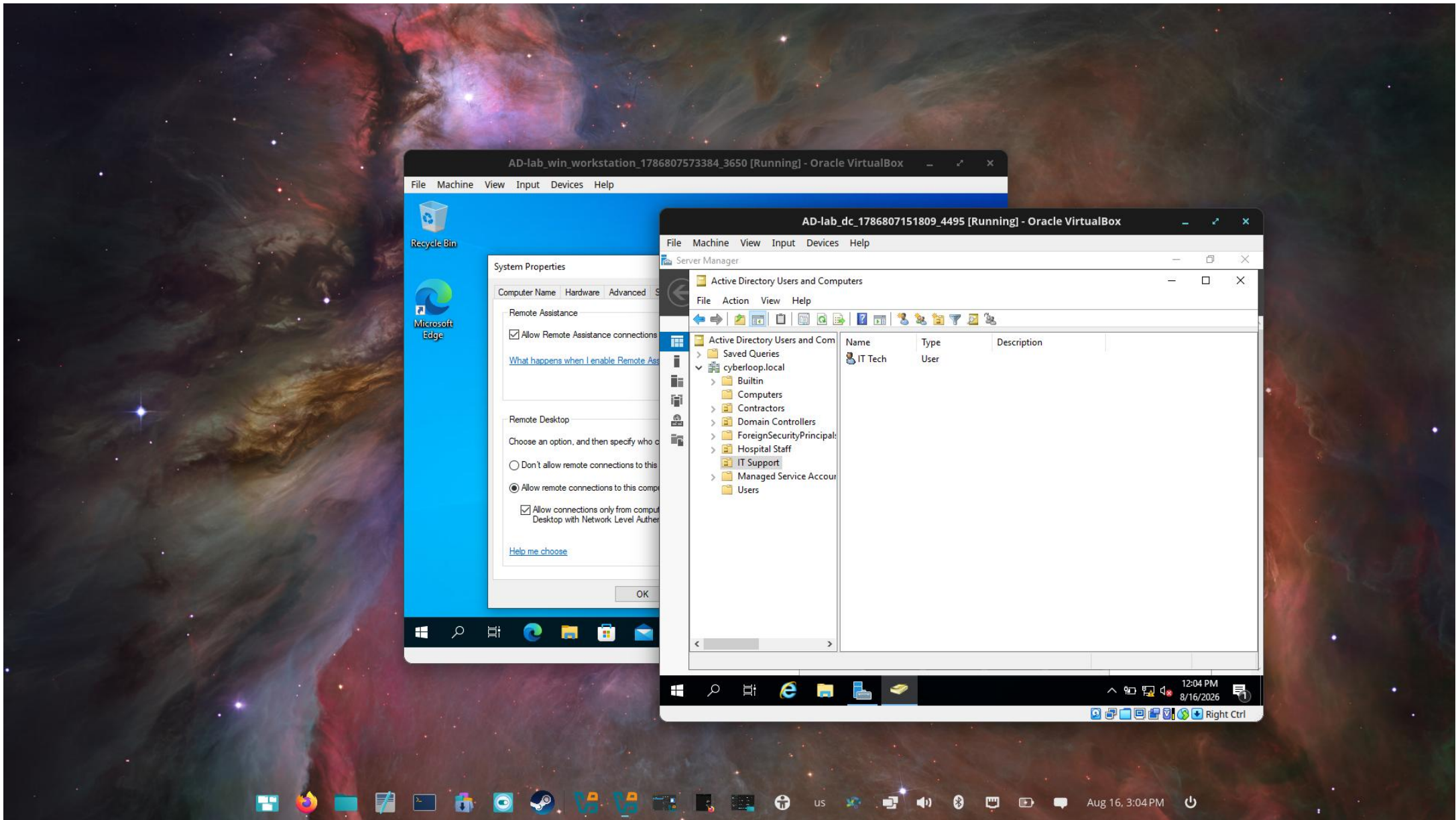
DOMAIN-CONTROLL 6008 Error EventLog System 8/15/2026 4:14:07 PM

DOMAIN-CONTROLL 34 Warning Disk System 8/15/2026 4:14:05 PM

DOMAIN-CONTROLL 34 Warning Disk System 8/15/2026 4:14:05 PM

5:13 PM 8/15/2026





AD-lab_win_workstation_1786807573384_3650 [Running] - Oracle VirtualBox

File Machine View Input Devices Help

Settings

Home

Find a setting

Accounts

- Your info
- Email & accounts
- Sign-in options
- Access work or school
- Sync your settings

Your info

IT TECH
CYBERLOOP\ittech

Create your picture

Camera

Browse for one

Sunday, August 16, 2026

11:56 AM
8/16/2026

Right Ctrl

07151809_4495 [Running] - Oracle VirtualBox

Manage Tools View Help

WELCOME TO SERVER MANAGER

- 1 Configure this local server
- 2 Add roles and features
- 3 Add other servers to manage
- 4 Create a server group
- 5 Connect this server to cloud services

Hide

SERVER GROUPS

Server groups: 1 | Servers total: 1

AD DS 1

DNS 1

11:56 AM
8/16/2026

Right Ctrl

File Edit View

os_pop@pop-os: ~/AD-lab

```
VM, run 'vagrant status NAME'.
os_pop@pop-os:~/AD-lab$ vagrant up win_workstation
Bringing machine 'win_workstation' up with 'virtualbox' provider...
==> win_workstation: Checking if box 'StefanScherer/windows_10' version '2021.12.09' is up to date...
==> win_workstation: Clearing any previously set forwarded ports...
==> win_workstation: Fixed port collision for 3389 => 3389. Now on port 2200.
==> win_workstation: Fixed port collision for 5986 => 55986. Now on port 2201.
==> win_workstation: Fixed port collision for 22 => 2222. Now on port 2202.
==> win_workstation: Clearing any previously set network interfaces...
==> win_workstation: Preparing network interfaces based on configuration...
win_workstation: Adapter 1: nat
win_workstation: Adapter 2: hostonly
==> win_workstation: Forwarding ports...
win_workstation: 3389 (guest) => 2200 (host) (adapter 1)
win_workstation: 3389 (guest) => 43389 (host) (adapter 1)
win_workstation: 5985 (guest) => 45985 (host) (adapter 1)
win_workstation: 5986 (guest) => 2201 (host) (adapter 1)
win_workstation: 22 (guest) => 2202 (host) (adapter 1)
==> win_workstation: Running 'pre-boot' VM customizations...
==> win_workstation: Booting VM...
==> win_workstation: Waiting for machine to boot. This may take a few minutes...
win_workstation: WinRM address: 127.0.0.1:45985
win_workstation: WinRM username: vagrant
win_workstation: WinRM execution_time_limit: PT2H
win_workstation: WinRM transport: negotiate
==> win_workstation: Machine booted and ready!
==> win_workstation: Checking for guest additions in VM...
win_workstation: The guest additions on this VM do not match the installed version of
win_workstation: VirtualBox! In most cases this is fine, but in rare cases it can
win_workstation: prevent things such as shared folders from working properly. If you see
```

```
File Edit View os_pop@pop-os: ~ + - ↗ ×
os_pop@pop-os: ~ × os_pop@pop-os: ~ × os_pop@pop-os: ~ ×
os_pop@pop-os:~$ ^[[200~xfreerdp /v:127.0.0.1:23389 /u:vagrant /p:vagrant /cert:ignore~
xfreerdp: command not found
os_pop@pop-os:~$ xfreerdp /v:127.0.0.1:23389 /u:vagrant /p:vagrant /cert:ignore
[15:14:45:414] [21445:21446] [ERROR][com.freerdp.core.transport] - BIO_should_retry returned a system error 32: Broken pipe
[15:14:45:414] [21445:21446] [ERROR][com.freerdp.core] - transport_write:freerdp_set_last_error_ex ERRCONNECT_CONNECT_TRANSPORT_FAILED [0x00020000]
[15:14:45:421] [21445:21446] [ERROR][com.freerdp.core.transport] - BIO_should_retry returned a system error 32: Broken pipe
[15:14:45:421] [21445:21446] [ERROR][com.freerdp.core] - transport_write:freerdp_set_last_error_ex ERRCONNECT_CONNECT_TRANSPORT_FAILED [0x00020000]
[15:14:45:421] [21445:21446] [ERROR][com.freerdp.core] - freerdp_post_connect failed
os_pop@pop-os:~$ xfreerdp /v:127.0.0.1:23389 /u:vagrant /p:vagrant /cert:ignore
[15:19:08:908] [22352:22353] [ERROR][com.freerdp.core.transport] - BIO_should_retry returned a system error 32: Broken pipe
[15:19:08:908] [22352:22353] [ERROR][com.freerdp.core] - transport_write:freerdp_set_last_error_ex ERRCONNECT_CONNECT_TRANSPORT_FAILED [0x00020000]
[15:19:08:921] [22352:22353] [ERROR][com.freerdp.core.transport] - BIO_should_retry returned a system error 32: Broken pipe
[15:19:08:921] [22352:22353] [ERROR][com.freerdp.core] - transport_write:freerdp_set_last_error_ex ERRCONNECT_CONNECT_TRANSPORT_FAILED [0x00020000]
[15:19:08:921] [22352:22353] [ERROR][com.freerdp.core] - freerdp_post_connect failed
os_pop@pop-os:~$
```